



Solution Set

Question 1

For

$$f(x, y) = (x^2 - 1)(y + 2)$$

hold y fixed for

$$f_x$$

and hold x fixed for

$$f_y$$

$$f_x = 2x(y + 2), \quad f_y = x^2 - 1$$

For

$$f(x, y) = e^{x+y+1}$$

each partial derivative is

$$f_x = e^{x+y+1}, \quad f_y = e^{x+y+1}$$

Question 2

The equation is

$$xy - yz - xz = 0$$

so solve explicitly for z :

$$xy = z(x + y) \Rightarrow z = \frac{xy}{x+y}$$

Then

$$z_x = \frac{y(x+y) - xy}{(x+y)^2} = \frac{y^2}{(x+y)^2}$$

$$z_y = \frac{x(x+y) - xy}{(x+y)^2} = \frac{x^2}{(x+y)^2}$$

Question 3

Write

$$z = \log \sqrt{x^2 + y^2} = \frac{1}{2} \log(x^2 + y^2)$$

Then

$$z_x = \frac{x}{x^2 + y^2}, \quad z_y = \frac{y}{x^2 + y^2}$$

Hence

$$xz_x + yz_y = \frac{x^2 + y^2}{x^2 + y^2} = 1$$

Question 4

For

$$f(x, y) = \log(x^2 + y^2)$$
$$f_x = \frac{2x}{x^2 + y^2}, \quad f_{xx} = \frac{2(y^2 - x^2)}{(x^2 + y^2)^2}$$
$$f_y = \frac{2y}{x^2 + y^2}, \quad f_{yy} = \frac{2(x^2 - y^2)}{(x^2 + y^2)^2}$$

Adding gives

$$f_{xx} + f_{yy} = 0$$

Question 5

Use the chain rule:

$$\frac{du}{dt} = u_x x'(t) + u_y y'(t) + u_z z'(t)$$

Here

$$u_x = 2x, \quad u_y = -4y, \quad u_z = 3z^2$$

and

$$x = \sin t, \quad x' = \cos t, \quad y = e^t, \quad y' = e^t, \quad z = 3t, \quad z' = 3$$

Therefore

$$\frac{du}{dt} = 2 \sin t \cos t - 4e^{2t} + 81t^2$$

Question 6

For

$$V(S_0, r, T) = S_0 - Ke^{-rT}$$

the first-order PnL for small changes in spot, rate, and time-to-maturity is

$$dV = V_S dS_0 + V_r dr + V_T dT$$

The partial derivatives are

$$V_S = 1, \quad V_r = KTe^{-rT}, \quad V_T = Kre^{-rT}$$

So the small-change PnL equation is

$$dV \approx dS_0 + KTe^{-rT} dr + Kre^{-rT} dT$$

If calendar time passes while expiry date is fixed, then

$$dT < 0$$

and the theta contribution enters through the sign of dT.